# ACT Bushfire Management Standards

**ACT Strategic Bushfire Management Plan Version 3**

**2014**

# Background

These standards are prepared under the ACT Strategic Bushfire Managements Plan Version 3. They detail the measurable outcomes required under the current and ongoing management policies and procedures detailed in Part Two of the plan. This document supports the Fire Services, land managers, developers as well as the general community in achieving effective results in reducing bushfire risk.

The ACT Emergency Services Commissioner is responsible for approval of these standards. They will be reviewed as required to reflect significant changes.

# Zoning and fuel management standards

## Part one – Determining the widths of zones

### Asset interface classification

A supplementary risk assessment is required for new developments and infill to specifically provide guidance for the management of consequences adjacent to built-up areas. Based on the past experience and fire weather analyses, it is an appropriate risk management decisions to vary the distance of the Asset Protection Zone buffers, reflecting the likelihood of flame intensity from a particular direction.

Zoning for these areas is determined by the Asset Interface Classification. This classification of primary, secondary and lee edges along the rural–urban edge is based on the level of bushfire risk the interface is exposed to. It considers:

- the maximum fire size an asset may be subject to
- the part of the fire (head, flank, back) an asset may be subject to, recognising the major fire threats from the north and west.
- the length of potential fire runs.

Table 1 details the mechanism by which this classification is defined. As appropriate, this classification may be modified for short distances to provide for consistency between adjacent classifications such as uphill (higher intensity) or downhill (lower intensity) fire runs to assets.

**Table 1** Asset Interface Classification matrix

| | Length of fire run to asset interface (metres) | | |
|---|---|---|---|
| **Aspect of fire run** | **<100** | **100–350** | **>350** |
| North | Secondary | Primary | Primary |
| Northest | Secondary | Primary | Primary |
| West | Secondary | Primary | Primary |
| Southwest | Lee | Secondary | Primary |
| South | Lee | Secondary | Secondary |
| Southeast | Lee | Lee | Lee |
| East | Lee | Lee | Secondary |
| Northeast | Lee | Lee | Secondary |

## Asset Protection Zones

APZ defined under this section relate to fuel managed areas on government managed lands. They may contribute to the requirement for APZ's under AS 3959.

Tables 2 and 3 establish APZ requirements. Table three provides requirements where no Outer APZ is implemented. These are minimum standards only and variations may be applied, for example where slope may significantly affect fire behaviour or to meet minimum requirements for AS 3959 assessments in Bushfire Prone Areas. Application of these zones shall be approved by the ESA

**Table 2 -** Widths for Inner and Outer Asset Protection Zones (APZs)

| Vegetation type | Asset Interface Classification (as mapped) | Inner APZ | Outer APZ |
|---|---|---|---|
| | | Width (m) | Width (m) |
| Forest and woodland | Primary | 30 | Target 300; minimum 200 |
| | Secondary | 20 | 100 |
| | Lee | 10 | 0 |
| Grass and open woodland | Primary | 30 | 100 |
| | Secondary | 20 | 0 |
| | Lee | 10 | 0 |

**Table 3 -** Width for Inner APZs (with no Outer APZ)

| Vegetation type | Asset Interface Classification (as mapped) | Inner APZ | Outer APZ |
|---|---|---|---|
| | | Width (m) | Minimum Width (m) |
| Forest and woodland | Primary | 60 | 0 |
| | Secondary | 40 | 0 |
| | Lee | 20 | 0 |
| Grass and open woodland | Primary | 40 | 0 |
| | Secondary | 20 | 0 |
| | Lee | 10 | 0 |

## Part two – Fuel management standards

Table 4 - Fuel management standards – Asset Protection Zones

| APZ Type | Vegetation type | Fuel management standards |
|---|---|---|
| Inner | | |
| Default standards to be applied across at least 80% of the zones as mapped. Where default standards cannot be achieved, the responsible land manager may identify alternative treatments to meet the overall objectives for the zone. Any significant variation on the default standards shall be approved by the ESA. | Forest and woodland | Maintained at an overall fuel hazard ≤low<br>3–5 m canopy separation or fuel gap to crown, > 3 m maintained |
| | Grass and open woodland | Grassland maintained at ≤200 mm height when grassland curing ≥70%. |
| Outer | | |
| Default standards to be applied across at least 70% of the zones as mapped. Where default standards cannot be achieved, the responsible land manager may identify alternative treatments to meet the overall objectives for the zone. Any significant variation on the default standards shall be approved by the ESA. | Forest and woodland | Overall fuel hazard ≤moderate |
| | Grass and open woodland | Grassland fire hazard[a] ≤35 when grassland curing ≥70% |

ESA = ACT Emergency Services Agency

a The grassland fire hazard is a score deriving from the height of grass (m) and the amount of cover (%). See Section 21.2.1, 'Explanatory notes'.

**Table 5 -** Fuel management standards – Strategic Firefighting Advantage Zones (SFAZs)

| SFAZ Type | Vegetation or break type | Fuel management standards |
|---|---|---|
| Broad area treatment | | |
| Application of broad-area hazard reduction burning that aims to reduce fuel across multiple landscape elements (ridges, side slopes, gullies) and treat at least 70% of the area identified.<br>Where default standards cannot be achieved, the responsible land manager may identify alternative treatments to meet the overall objectives for the zone. Any significant variation on the default standards shall be approved by the ESA. | Forest and shrubland | Overall fuel hazard ≤ high |
| | Grass and open Woodland | Grassland fire hazard ≤50 when grassland curing ≥70%. |
| | Identified arterial roads, rural roads and easements | Grassland fire hazard ≤35 when grassland curing ≥70%. |
| Linear breaks | | |
| Standards to be applied across at least 70% of the zones as mapped. Where default standards cannot be achieved, the responsible land manager may identify alternative treatments to meet the overall objectives for the zone. | Identified arterial roads, rural roads and easements | Grassland fire hazard ≤35 when grassland curing ≥70%. |
| | Forest greenbreaks | Overall fuel hazard ≤ high |
| Ridgetop burning treatments | | |
| Application of broad-area hazard reduction burning that aims to reduce fuel, primarily along ridgelines and side slopes aiming to treat in the order of 30–50% of the gross area identified as 'ridgetop burns' | n/a | Overall fuel hazard ≤ high |
| Plantations | | |
| | Immature stands (<15 years old): | Pruned to 2.0 m and thinned to 650 sph nominally at Year 8-10. Debris from thinning and pruning to be <1 m high. In areas immediately adjacent to potential ignition sources or built assets consider mechanical crushing of thinning and pruning debris or burning as soon as practicable without damaging the residual stand. |
| | Mature stands (>15 years old) | Mature stands (>15 years old). Schedule harvesting to reduce contiguous areas of untreated slash during the fire season. Remove the outside row of trees (adjacent to fire trails) during first commercial thinning operation to improve access on fire trails. Maintain strategic fuel breaks through the maintenance of road pavements and verges, and edge pruning and thinning. |

ESA = ACT Emergency Services Agency

**Table 6 -** Fuel management standards – other zones

| Zone | Treatment standards |
|---|---|
| Landscape Fire Management Zone | Fuel management standards not applied. |
| Agricultural Fire Management Zone | Requirements will be defined in bushfire operational plans, developed through the Farm Firewise program. |

## Explanatory notes – fuel management standards

*Overall fuel hazard*

Overall fuel hazard assessment is currently consistent with the Territory and Municipal Services Fuel Assessment Methodology. This encompasses the processes identified in:

- *Overall fuel hazard guide*, 3rd edition, Victorian Department of Sustainability and Environment, May 1999
- Project Vesta. Fire in dry eucalypt forests: fuel structure, fuel dynamics and fire behaviour, JS Gould, WL McCaw, NP Cheney, PF Ellis, IK Knight and AL Sullivan, CSIRO/SCION, 2007

The process used for the assessment of bushfire fuels is included in Part one of the supporting information. Subsequent revisions or changes to methodologies may be applied as appropriate.

*Grassland curing*

Assessment of grassland curing is currently consistent with the *Victorian Country Fire Authority grassland curing guide* (CFA, 1999). Subsequent revisions or changes to methodologies may be applied as appropriate.

*Grassland fuel hazard*

A combination of height and cover is currently used to determine triggers for short-to-medium term treatments when grasses are sufficiently cured to carry free-burning fires (approximately 70% cured). Although significant variation exists, grass height (m) and cover (%) are related to fuel load and visibility, which are important factors affecting the ability to suppress grassfires. A score (the Grassland Fuel Hazard – see Table 21.8) was developed for the ACT and derived by multiplying these two factors, which provides a means of defining allowable fuel conditions in Outer Asset Protection Zones and Strategic Firefighting Advantage Zones. Subsequent revisions or changes to methodologies may be applied as appropriate.

**Table7-** Grassland Fuel Hazard scores

| Height (m) | Cover (%) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | **10** | **20** | **30** | **40** | **50** | **60** | **70** | **80** | **90** | **100** |
| 0.1 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 0.2 | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 |
| 0.3 | 3 | 6 | 9 | 12 | 15 | 18 | 21 | 24 | 27 | 30 |
| 0.4 | 4 | 8 | 12 | 16 | 20 | 24 | 28 | 32 | 36 | 40 |
| 0.5 | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 |
| 0.6 | 6 | 12 | 18 | 24 | 30 | 36 | 42 | 48 | 54 | 60 |
| 0.7 | 7 | 14 | 21 | 28 | 35 | 42 | 49 | 56 | 63 | 70 |
| 0.8 | 8 | 16 | 24 | 32 | 40 | 48 | 56 | 64 | 72 | 80 |
| 0.9 | 9 | 18 | 27 | 36 | 45 | 54 | 63 | 72 | 81 | 90 |
| 1.0 | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |

# Classification and recommended standards for access

Sources: Australian Capital Territory (ACT) *Strategic Bushfire Management Plan* (SBMP), version 2, 2009; and New South Wales Rural Fire Service P*lanning for bushfire protection*, 2006.

## Ground access standards

Fire access includes a range of vehicular tracks, roads and walking tracks on public land, access trails on rural leases, and the public road network in the urban and rural areas of the ACT. These features reduce the response time to fires and make it safer to undertake bushfire suppression operations.

Fire trails, and the public roads and walking tracks also provide potential control lines for containing bushfires and for conducting prescribed burns.

### Standards – fire trails

The classification of fire roads, tracks and trails (Table 8) is performance based to provide clear guidance to land managers and response agencies during incidents.

Table 8 identifies the requirements for providing passing opportunities on fire trails, and requirements for providing vehicle turnaround capability on dead-end trails.

In grassland areas or in rural areas or reserves, the mineral earth ground fire access network may be supplemented by slashed grass breaks or ad hoc access trails through properties or reserves. These breaks provide additional vehicle access during suppression operations, but do not provide as safe suppression areas as mineral earth roads. These slashed trails are generally not mapped and the locations may change over time.

**Table 8 -** Fire trail access standards

| Classification | Performance standard | Nominal width and maximum grade | Surface | Nominal horizontal curve radius | Nominal vegetation maintenance |
|---|---|---|---|---|---|
| Walking track | To allow the safe passage of firefighters on foot | Generally these features will be less than 1 m in width | Natural surface, grass, sealed or gravel surface | Not applicable | Vegetation maintained to allow single-file walking |
| Light unit | To allow the safe passage of light units (e.g. 4 × 4 Landcruiser or similar, carrying a water tank of 400–600 litres) | Generally these features will have a width of 3 m, with a maximum grade of 20 degrees | Natural surface, grass, sealed or gravel surface | Corners of sufficient radius to make 3-point turns by light units unnecessary | Roadside vegetation maintained to allow unimpeded access by light units |
| Tanker | To allow the safe passage of tankers (e.g. 4 × 4 trucks, 8–12 tonnes, carrying a water tank of 2500–5000 litres) | Generally these features will have a width of 4 m, with a maximum grade of 15 degrees | Natural surface, grass, sealed or gravel surface | Corners of sufficient radius to make 3-point turns by tankers unnecessary | Roadside vegetation maintained to allow unimpeded access by tankers |
| Rigid float | To allow the safe passage of a truck, including large CAFS tankers and large rural tankers with a rigid body carrying a small (D3- or D4-sized) dozer | Generally these features will have the same characteristics as tanker roads, but will have smaller rollovers, no splash creek crossings and no switchback corners. | | | |
| Articulated float | To allow the safe passage of articulated floats (truck and trailer carrying a D6-sized dozer) | Generally these features will have a width of 5 m and a maximum grade of 12 degrees | Natural surface, grass, sealed or gravel surface | Corners of sufficient radius to make 3-point turns by a float unnecessary | Roadside vegetation maintained to allow unimpeded access by an articulated float |

CAFS = compressed air foam system

**Table 9-** Nominal requirements for passing and dead end fire trails

| | |
|---|---|
| **Passing opportunities** | Use road verges and natural areas wherever possible to allow two vehicles to safely pass.<br>In the case of light unit trails, to allow two light units to pass, and in the case of tanker trails, to allow two tankers to pass.<br><br>If there are no opportunities for vehicles to pass in a 250 m section, where terrain and ecological constraints allow, provide for widening of pavement, verges or the construction of passing bays. Where dormant trails are reopened during suppression operations, suitable passing areas should be constructed where terrain and ecological constraints allow to ensure vehicles can pass within a 250 m section. |
| **Dead ends** | Avoid as far as possible and signpost where they occur.<br><br>Where dead ends exist, a turnaround of sufficient radius for the standard of the trail must be provided (e.g. in the case of light unit trails, the turnaround must allow a light unit to turn around on full lock).<br><br>If there is insufficient space for such a turnaround due to the topography or other constraints, provision should be made to allow a maximum three-point turn or a 'T' or 'Y' shaped turnaround area. The dimensions of the 'T' or 'Y' shaped turn around must be appropriate for the type of vehicles that will use the trail.<br><br>Turnarounds should be constructed when dormant trails are reconstructed for suppression operations, and temporary signage will be provided at the start of these trails. |

## Standards – Public Roads

**Table 10 -** Access standards for public roads for new estate development

| Performance criteria requirement | Acceptable solutions |
|---|---|
| Firefighters are provided with safe all-weather access to Bushfire Prone Areas and assets. | Refer to ESDD Estate Development Code March 2012<br><br>Public roads are two-wheel drive, all-weather roads. |
| Public road widths and design allow safe access and egress for firefighters while residents are evacuating an area.<br>Public road widths allow firefighting crews to work with firefighting equipment around the vehicle, and to allow other vehicles to pass with safety. | Edge road required for all new subdivisions and developments. Alternate solutions will be considered on merits of safety to public and emergency service personnel.<br>Urban edge roads are two-way – that is, at least two traffic lane widths (carriageway 7.5 m minimum kerb to kerb), allowing traffic to pass in opposite directions, with parking provided in designated parking bays clear of the carriageway.<br><br>Hydrants are located clear of parking bays.<br>The edge road is linked to the internal road system at an interval of no greater than 500 m in urban areas.<br>Traffic management devices are constructed to facilitate access and egress by emergency services vehicles.<br>Public roads have a cross-fall not exceeding 6%. All roads are through roads. Dead-end roads are not recommended but, if unavoidable, dead ends are not more than 200 m in length, incorporate a minimum 24-m-diameter unobstructed turning circle, and are clearly signposted as a dead end and direct traffic away from the hazard.<br><br>Curves of roads (other than perimeter roads) are a minimum inner radius of 6 m and minimal in number, to allow for rapid access and egress.<br><br>The minimum distance between inner and outer curves is 6 m.<br><br>Maximum grades for sealed roads do not exceed 28%, and an average grade of not more than 18% or other gradient specified by road design standards, whichever is the lesser gradient.<br><br>There is a minimum vertical clearance to a height of 4.2 m above the road at all times.<br><br>Roads are clearly signposted (with easily distinguishable names), and buildings and properties are clearly numbered.<br><br>In designated Bushfire Prone Areas, cul-de-sac road design is generally not encouraged. Where they are used, however, they should not exceed 200 m in length. In some instances, it may be possible to provide emergency access between cul-de-sac heads so that residents and firefighters have two-way access and egress. In this case, and provided it does not service more than 8 lots, the maximum length of a cul-de-sac can be increased to 600 m. Turnaround areas should allow fire appliances to turn around safely and should be available at cul-de-sac heads, house sites and at 250 m intervals along driveways and fire service accesses.<br><br>Emergency accesses may be used to link up with roads to allow alternative access and egress during emergencies where traffic flow designs do not allow for two-way access. The access should comply with minimum standards for roads and should be signposted. If gates are used to control traffic flow during non-emergency periods, they must not be locked. |
| The capacity of road surfaces and bridges is sufficient to | The capacity of road surfaces and bridges is sufficient to carry fully loaded firefighting vehicles (approximately 30 tonnes for aerial appliances and 25 tonnes for tankers.). Bridges shall be signposted to |

| | |
|---|---|
| carry fully loaded firefighting vehicles. | clearly indicate load rating. |
| There is clear access to reticulated water supply. | Internal public roads 6.5 m wide provide parking within parking bays, and locate services outside of the parking bays to ensure accessibility to reticulated water for fire suppression.<br><br>Internal public roads between 6.5 m and 7.5 m are signposted as 'No Parking' on one side, with the services (hydrants) located on this side to ensure accessibility to reticulated water for fire suppression.<br><br>One-way only public access roads are no less than 3.5 m wide and provide parking within parking bays, and locate services outside of the parking bays to ensure accessibility to reticulated water for fire suppression. |
| Parking does not obstruct the minimum paved width. | Parking bays are a minimum of 2.6 m wide from kerb edge to road pavement. No services or hydrants are located within the parking bays.<br><br>Public roads directly interfacing the bushfire hazard vegetation shall provide roll-top kerbing to the hazard side of the road. |

# Aerial access

## Aerial access classification

Helipads are natural or constructed features on which helicopters may safely land. Helipads provide access for firefighters to suppress remote area fires, and locations from which helicopters may operate during bushfire suppression and prescribed burning operations. Helipads are broadly classified according to the size of the helicopter that may safely land on them, although the decision to use a helipad is ultimately rests with the pilot in charge of the aircraft. Indicative helipad standards are at Table 22.4.

Within the ACT, there are a number of strategically advantageous helipads that have been identified and constructed to provide rapid access to remote areas in circumstances where there is no other viable method of access. These permanent helipads (including suitable natural features) have been identified and classified as per Table 11. Additional temporary helipads may be constructed to facilitate enhanced access to a remote area during a bushfire operation.

**Table 11 -** Helipad standards

| Classification | Nominal description |
|---|---|
| Light | To allow the safe landing of light helicopters (e.g. Jet Rangers, AS350BAs, BK117). These helipads will usually consist of the following three zones:<br>• Zone A – a rock/grassed/mineral earth area 7 x 7 m with no protrusions such as tree stumps or rocks above 20 cm in height<br>• Zone B – an area cleared of trees and tall shrubs for an additional 13 m with no large protrusions such as tree stumps or large rocks above a height of 0.4 m<br>• Zone C – all tall trees outside of Zone B removed which may obstruct an approach angle of 40 degrees to Zone A, in one or more directions. |
| Medium | To allow the safe landing of medium-sized helicopters (e.g. Bell 412). These helipads will usually consist of the following three zones:<br>• Zone A – a rock/grassed/mineral earth area 15 x 15 m with no protrusions such as tree stumps or rocks above 20 cm in height<br>• Zone B – an area cleared of trees and tall shrubs for an additional 20 m with no large protrusions such as tree stumps or large rocks above a height of 0.4 m<br>• Zone C – All tall trees outside of Zone B removed which may obstruct an approach angle of 40 degrees to Zone A, in one or more directions. |

Where no other access method is viable, or when deemed operationally necessary, remote area firefighters may access an area by being winched in by helicopter; however, this type of operation is considerably riskier than using a helipad, and is avoided where possible. After being inserted via a winching operation, remote area firefighters will normally construct a temporary helipad close to the

bushfire to allow helicopters to safely land, removing the need to undertake further winching operations.

### Aerial access construction and maintenance

Permanent helipads must be constructed and maintained to a suitable standard to ensure a safe working environment and ameliorate social, economic and environmental impacts. Existing natural clearings should be used wherever practicable.

New permanent helipads will only be constructed where they provide a clear strategic benefit and the financial and environmental costs of construction and maintenance do not outweigh the benefits for fire management. In certain circumstances, temporary helipads constructed as part of a bushfire suppression activity may be maintained as new permanent helipads where they also meet these criteria.

The use of helipads during fire suppression operations must consider the impacts on hydrological and environmental values. The transport and storage of fuel and fire retardants on or adjacent to helipads in remote locations must ensure that measures are taken to avoid spillage or contamination of the site. In some cases this will not be permitted.

Existing helipads (including suitable natural features) will be mapped and classified by TAMSD and provided to the ESA.

### Road sign standards

Sufficient roads and tracks will be signposted to allow navigation through the fire trail network (note that this may not require all roads and trails to be signposted). A systematic program of phasing in new signs on identified fire access will be implemented, consistent with resource availability.

## Utility services in urban and rural areas

Source: NSW Rural Fire Service, *Planning for bushfire protection,* 2006.

**Table 12 -** Standards for utility services in urban and rural areas

| Performance criteria | Acceptable solutions |
|---|---|
| **Reticulated water supplies** | |
| Water supplies are easily accessible and located at regular intervals | The Water and Sewerage Network (Design and Maintenance) Code of the *Utilities Act 2000* requires the fire-fighting requirements are able to be met.<br><br>A deed of agreement exists between ACTEW Corporation Limited and ACTF&R in relation to water supply in the built up area. This agreement details operative provisions which cover:<br>• Fire Hydrants – general provisions<br>• Flow rates<br>• Fire risk classification and fire hydrant spacings<br>• Fire hydrant testing and maintenance<br>• Fire hydrant system shutdown / isolation<br>• Connection to domestic supplies<br>• Water usage by ACTF&R and ACTRFS<br>• Provision of plans showing location of fire h<br>• ydrants on the water network<br>• Amendments to water supply standards<br>The deed of agreement is currently under review by both parties. |
| | |
| For rural and rural village developments (or settlements) in Bushfire Prone Areas, a water supply reserve dedicated to firefighting purposes is installed and maintained. The water supply can be an amalgam of minimum quantities for each subdivision lot (community-titled subdivisions) or held individually on each lot | Requirements for rural villages and properties should refer to Planning for Bushfire Protection Chapter 4 table 4.2. |
| **Gas** | |
| Location and installation of gas services will not lead to ignition of surrounding bushland or the fabric of buildings | As detailed in AS 3959 which notes water and gas pipes above ground.<br><br>Exposed water and gas supply pipes shall be metal. |

## Emergency and evacuation planning in Bushfire Prone Areas

Source: NSW Rural Fire Service, *Planning for bushfire protection*, 2006.

**Table13 -** Standards for emergency and evacuation planning in Bushfire Prone Areas

| Performance criteria | Acceptable solutions |
|---|---|
| An emergency and evacuation management plan shall be prepared for Special Fire Protection Developments | Compliance with AS 3745-2002 'Emergency control organisation and procedures for buildings, structures and workplaces' for residential accommodation'.<br>Compliance with AS 4083-1997 'Planning for emergencies - for health care facilities'.<br><br>(Domestic dwellings within the BPA should refer to the ESA website) |